

# WIRING SUMMARY

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## Arduino UNO → L298N Motor Driver

| Arduino Pin | L298N Pin | Function                   |
|-------------|-----------|----------------------------|
| D2          | IN1       | Left motor direction       |
| D6          | IN2       | Left motor direction       |
| D7          | IN3       | Right motor direction      |
| D8          | IN4       | Right motor direction      |
| D9          | ENA       | Left motor speed (enable)  |
| D10         | ENB       | Right motor speed (enable) |
| GND         | GND       | Common ground              |

## Arduino UNO → Ultrasonic Sensor (HC-SR04)

### Arduino Pin Sensor Pin

|     |      |
|-----|------|
| D11 | Trig |
| D12 | Echo |
| 5V  | VCC  |
| GND | GND  |

## Arduino UNO → Servo Motor

| Arduino Pin | Servo Wire             |
|-------------|------------------------|
| D3          | Signal (orange/yellow) |
| 5V          | Power (red)            |
| GND         | Ground (brown/black)   |

## L298N → Motors & Power

- OUT1, OUT2 → Left side DC motors (2x)
- OUT3, OUT4 → Right side DC motors (2x)
- 12V input terminal → Battery pack (2x 3.7V Li-ion, ~7.4V total)
- 5V output (if jumper is on) can optionally power the Arduino, or keep Arduino powered separately via USB/barrel jack